

What is NGINX?

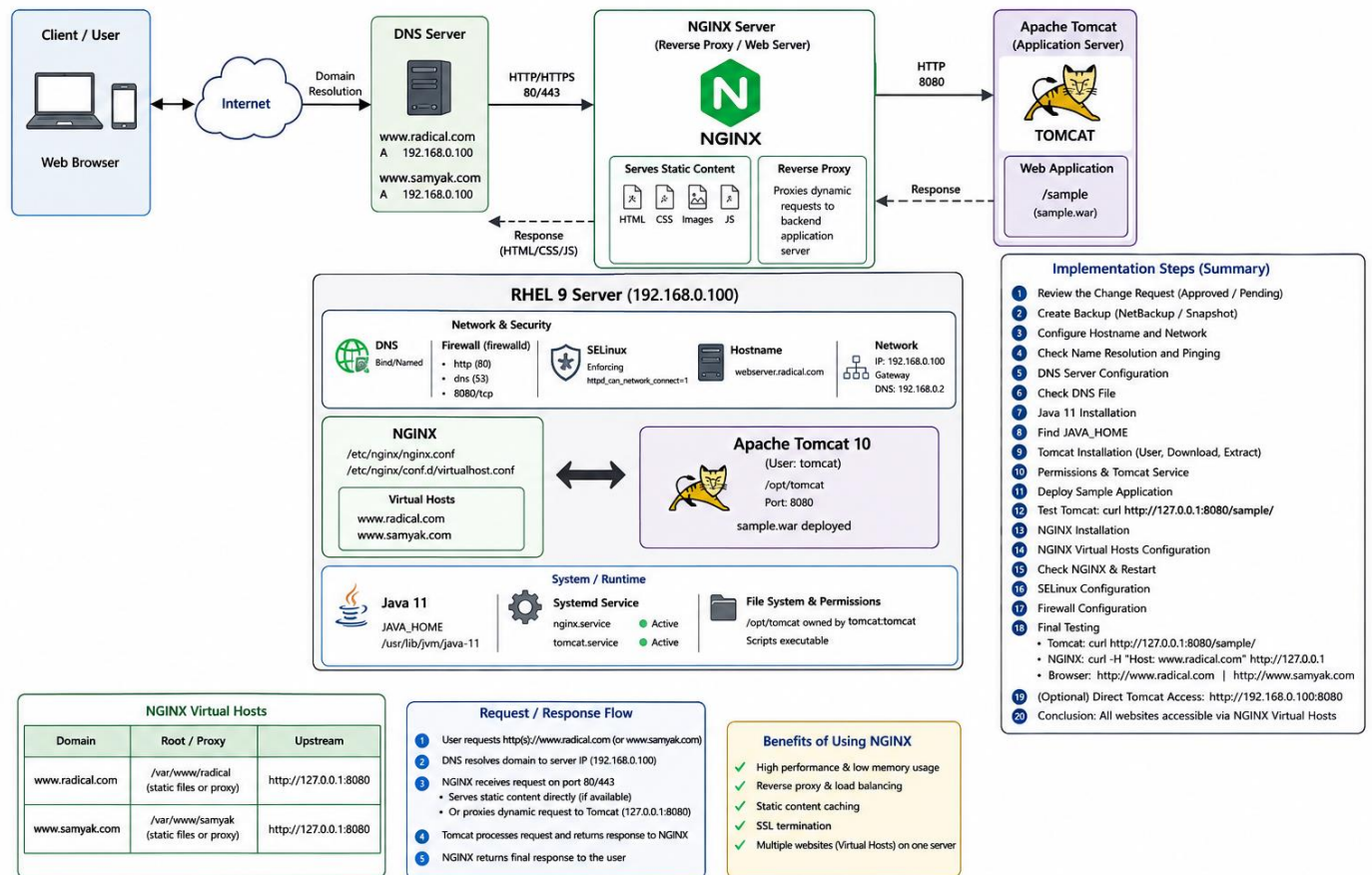
NGINX is a high-performance web server, reverse proxy server, load balancer, and HTTP cache. It is widely used to serve websites and applications efficiently.

```
graph TD; Client --> V1(( )); V1 --> Radical[www.radical.com]; Radical --> V2(( )); V2 --> NGINX["NGINX (192.168.0.100:80)"]; NGINX --> V3(( )); V3 --> Tomcat["Tomcat (192.168.0.100:8080)"]; Tomcat --> V4(( )); V4 --> App[sample.war Application];
```

The diagram illustrates the request flow in a web application architecture. It starts with a 'Client' sending a request, indicated by a downward arrow. Below the client is a vertical line, followed by a downward arrow pointing to the domain 'www.radical.com'. Another vertical line and downward arrow lead to 'NGINX (192.168.0.100:80)'. This is followed by another vertical line and downward arrow pointing to 'Tomcat (192.168.0.100:8080)'. A final vertical line and downward arrow lead to the 'sample.war Application'.

1. User sends a request to a website.
2. NGINX receives the request on port 80 (HTTP) or 443 (HTTPS).
3. NGINX either:
 - Serves static content directly (HTML, CSS, images), or
 - Forwards the request to a backend application server such as Apache Tomcat.
4. The backend processes the request and sends the response to NGINX.
5. NGINX returns the response to the user.

NGINX Architecture with Tomcat (RHEL 9)



1.Review the Change Request:

Confirm that the change request has been assigned to you and verify its current status (Approved or Pending). Proceed with the implementation only after receiving approval.

2.Create a Backup:

Before making any modifications, ensure a backup is taken. Use NetBackup for physical servers and create a snapshot for virtual machines to safeguard against potential issues.

3.Configure Hostname and network:-

```
[root@ngnix ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.100 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::20c:29ff:fe8c:cf33 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:8c:cf:33 txqueuelen 1000 (Ethernet)
    RX packets 484638 bytes 711822923 (678.8 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 77026 bytes 4577121 (4.3 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 363 bytes 34668 (33.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 363 bytes 34668 (33.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[root@ngnix ~]# hostname
ngnix.radical.com
[root@ngnix ~]#
```

4.Check name resolution and pinging :-

```
[root@ngnix ~]# ping www.redhat.com
PING e3396.dscx.akamaiedge.net (23.211.156.107) 56(84) bytes of data.
 54 bytes from a23-211-156-107.deploy.static.akamaitechnologies.com (23.211.156.107): icmp_seq=1 ttl=128 time=8.41 ms
 54 bytes from a23-211-156-107.deploy.static.akamaitechnologies.com (23.211.156.107): icmp_seq=2 ttl=128 time=10.7 ms
 54 bytes from a23-211-156-107.deploy.static.akamaitechnologies.com (23.211.156.107): icmp_seq=3 ttl=128 time=9.45 ms
 54 bytes from a23-211-156-107.deploy.static.akamaitechnologies.com (23.211.156.107): icmp_seq=4 ttl=128 time=9.46 ms
^C
--- e3396.dscx.akamaiedge.net ping statistics ---
 4 packets transmitted, 4 received, 0% packet loss, time 3005ms
 rtt min/avg/max/mdev = 8.410/9.501/10.683/0.804 ms
[root@ngnix ~]# ping www.google.com
PING www.google.com (142.251.154.119) 56(84) bytes of data.
 54 bytes from 142.251.154.119 (142.251.154.119): icmp_seq=1 ttl=128 time=9.19 ms
 54 bytes from 142.251.154.119 (142.251.154.119): icmp_seq=2 ttl=128 time=38.7 ms
 54 bytes from 142.251.154.119 (142.251.154.119): icmp_seq=3 ttl=128 time=9.93 ms
^C
--- www.google.com ping statistics ---
 3 packets transmitted, 3 received, 0% packet loss, time 2003ms
 rtt min/avg/max/mdev = 9.194/19.258/38.655/13.718 ms
[root@ngnix ~]# ping ngnix.radical.com
PING ngnix.radical.com (192.168.0.100) 56(84) bytes of data.
 54 bytes from ngnix.radical.com (192.168.0.100): icmp_seq=1 ttl=64 time=0.091 ms
 54 bytes from ngnix.radical.com (192.168.0.100): icmp_seq=2 ttl=64 time=0.131 ms
 54 bytes from ngnix.radical.com (192.168.0.100): icmp_seq=3 ttl=64 time=0.151 ms
 54 bytes from ngnix.radical.com (192.168.0.100): icmp_seq=4 ttl=64 time=0.102 ms
^C
--- ngnix.radical.com ping statistics ---
 4 packets transmitted, 4 received, 0% packet loss, time 3085ms
 rtt min/avg/max/mdev = 0.091/0.118/0.151/0.023 ms
[root@ngnix ~]#
```

5.DNS Server Configuration:-

#dnf install bind bind-utils -y

#systemctl enable --now named

#vi /etc/named.conf

```
erpris [root@ngnix tmp]# cat /etc/named.conf
//
erpris // named.conf
ig //
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
    listen-on port 53 { any; };
    listen-on-v6 port 53 { any; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secroots";
    recursing-file "/var/named/data/named.recursing";
    allow-query { any; };
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";

#####Setiing done by samyak#####
zone "radical.com" IN {
    type master;
    file "forward.zone";
};

zone "0.168.192.in-addr.arpa" IN {
    type master;
    file "reverse.zone";
};
[root@ngnix ~]#
```

#vi /var/named/forward.zone

#vi /var/named/reverse.zone

```
x  rhel 9  vsftpd testing  Nginx  x
[ root@nginx ~ ]# cat /var/named/forward.zone
$TTL 86400
@ IN SOA dnssrv.radical.com. root.radical.com. (
2026060101
3600
1800
604800
86400 )

@      IN NS   dnssrv.radical.com.

dnssrv IN A    192.168.0.100
nginx  IN A    192.168.0.100

www.radical IN A 192.168.0.100
www.samyak  IN A 192.168.0.100
[ root@nginx ~ ]# cat /var/named/reverse.zone
$TTL 86400
@ IN SOA dnssrv.radical.com. root.radical.com. (
2026060101
3600
1800
604800
86400 )

@      IN NS   dnssrv.radical.com.

100     IN PTR  nginx.radical.com.
[ root@nginx ~ ]#
```

6.Check DNS file :-

#named-checkconf

#named-checkzone radical.com /var/named/forward.zone

#named-checkzone 0.168.192.in-addr.arpa
/var/named/reverse.zone

```
[root@ngnix ~]# named-checkconf
[root@ngnix ~]# named-checkzone radical.com /var/named/forward.zone
zone radical.com/IN: loaded serial 2026060101
OK
[root@ngnix ~]# named-checkzone 0.168.192.in-addr.arpa /var/named/reverse.zone
zone 0.168.192.in-addr.arpa/IN: loaded serial 2026060101
OK
[root@ngnix ~]#
```

#systemctl restart named

#firewall-cmd --permanent --add-service=dns

#firewall-cmd --reload

7.Java 11 Installation:-

#dnf install java-11-openjdk java-11-openjdk-devel -y

#java -version

```
[root@ngnix ~]#
[root@ngnix ~]# dnf info java-11-openjdk
Updating Subscription Management repositories.
Last metadata expiration check: 0:32:12 ago on Mon 01 Jun 2026 02:32:36 PM IST.
Available Packages
Name       : java-11-openjdk
Epoch     : 1
Version    : 11.0.25.0.9
Release    : 7.el9
Architecture: x86_64
Size       : 435 k
Source     : java-11-openjdk-11.0.25.0.9-7.el9.src.rpm
Repository : rhel-9-for-x86_64-appstream-rpms
Summary    : OpenJDK 11 Runtime Environment
URL        : http://openjdk.java.net/
License    : ASL 1.1 and ASL 2.0 and BSD and BSD with advertising and GPL+ and GPLv2 and GPLv2 with exceptions and IJG and LGPLv3+ and MPLv2.0 and Public Domain and W3C and zlib and ISC and FTL and RSA
Description : The OpenJDK 11 runtime environment.

[root@ngnix ~]#
[root@ngnix ~]# dnf install java-11-openjdk java-11-openjdk-devel -y
Updating Subscription Management repositories.
Last metadata expiration check: 0:32:24 ago on Mon 01 Jun 2026 02:32:36 PM IST.
Dependencies resolved.
=====
Package                        Architecture    Version                               Repository
=====
Installing:
java-11-openjdk                x86_64          1:11.0.25.0.9-7.el9                 rhel-9-for-x86_64-appstream-rpms
java-11-openjdk-devel          x86_64          1:11.0.25.0.9-7.el9                 rhel-9-for-x86_64-appstream-rpms
Installing dependencies:
copy-jdk-configs               noarch          4.0-3.el9                           AppStream
=====
```

8. Check JAVA:-

```
#readlink -f $(which java)
```

What above command do- which java → finds Java command &
readlink -f → finds real path

```
#dirname $(dirname $(readlink -f $(which java)))
```

```
[root@ngnix ~]# readlink -f $(which java)
/usr/lib/jvm/java-11-openjdk-11.0.25.0.9-7.el9.x86_64/bin/java
[root@ngnix ~]# dirname $(dirname $(readlink -f $(which java)))
/usr/lib/jvm/java-11-openjdk-11.0.25.0.9-7.el9.x86_64
```

9.Tomcat Installation:-

Create tomcat user:

```
#useradd -r -m -U -d /opt/tomcat -s /bin/false tomcat
```

```
#cd /tmp
```

Download files from the internet-

```
# wget https://archive.apache.org/dist/tomcat/tomcat-10/v10.1.44/bin/apache-tomcat-10.1.44.tar.gz
```

```
wget https://archive.apache.org/dist/tomcat/tomcat-10/v10.1.44/bin/apache-tomcat-10.1.44.tar.gz
--2026-06-01 15:13:58-- https://archive.apache.org/dist/tomcat/tomcat-10/v10.1.44/bin/apache-tomcat-10.1.44.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14115940 (13M) [application/x-gzip]
Saving to: 'apache-tomcat-10.1.44.tar.gz'

apache-tomcat-10.1.44.tar.gz      100%[=====>] 13.46M
KB/s    in 2m 15s

2026-06-01 15:16:14 (102 KB/s) - 'apache-tomcat-10.1.44.tar.gz' saved [14115940/14115940]

[root@ngnix tmp]# ls -lh apache-tomcat-10.1.44.tar.gz
-rw-r--r--. 1 root root 14M Aug  4 2025 apache-tomcat-10.1.44.tar.gz
```


10.EXTRACT TOMCAT:-

```
# mkdir -p /opt/tomcat
```

```
#tar -xzf apache-tomcat-10.1.44.tar.gz \  
-C /opt/tomcat --strip-components=1
```

Check (fetch data from a URL)-

```
#curl https://downloads.apache.org/tomcat/tomcat-10/ | grep v10
```

```
[root@ngnix tmp]# curl https://downloads.apache.org/tomcat/tomcat-10/ | grep v10
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100  741  100  741    0     0   966      0 --:--:-- --:--:-- --:--:--   966
 <a href="v10.1.55/">v10.1.55/</a>                2026-05-11 21:20   -
[root@ngnix tmp]# cd /tmp

wget https://archive.apache.org/dist/tomcat/tomcat-10/v10.1.44/bin/apache-tomcat-10.1.44.tar.gz
--2026-06-01 15:13:58-- https://archive.apache.org/dist/tomcat/tomcat-10/v10.1.44/bin/apache-tomcat-10.1.44.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14115940 (13M) [application/x-gzip]
Saving to: 'apache-tomcat-10.1.44.tar.gz'

apache-tomcat-10.1.44.tar.gz      100%[=====] 13.46M  66.
KB/s    in 2m 15s

2026-06-01 15:16:14 (102 KB/s) - 'apache-tomcat-10.1.44.tar.gz' saved [14115940/14115940]
```

11. Set Permissions:-

```
#chown -R tomcat:tomcat /opt/tomcat
```

```
#chmod +x /opt/tomcat/bin/*.sh
```

12.Create Tomcat Service:-

#vi /etc/systemd/system/tomcat.service (editing a systemd service file for Tomcat)

```
[root@engnix tmp]# cat /etc/systemd/system/tomcat.service
[Unit]
Description=Apache Tomcat
After=network.target

[Service]
Type=forking

User=tomcat
Group=tomcat

Environment="JAVA_HOME=/usr/lib/jvm/java-11-openjdk-11.0.25.0.9-7.el9.x86_64"

Environment="CATALINA_HOME=/opt/tomcat"
Environment="CATALINA_BASE=/opt/tomcat"
Environment="CATALINA_PID=/opt/tomcat/temp/tomcat.pid"

ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh

Restart=on-failure

[Install]
WantedBy=multi-user.target
[root@engnix tmp]# systemctl daemon-reload
[root@engnix tmp]# systemctl enable --now tomcat
[root@engnix tmp]#
```

Start Tomcat:--

#systemctl daemon-reload

#systemctl enable --now tomcat

13.Deploy Sample Application:-

#cp /opt/tomcat/webapps/docs/appdev/sample/sample.war \

/opt/tomcat/webapps/

chown tomcat:tomcat /opt/tomcat/webapps/sample.war

#systemctl restart tomcat

```
[root@ngnix tmp]# cp /opt/tomcat/webapps/docs/appdev/sample/sample.war /opt/tomcat/webapps/
[root@ngnix tmp]# chown tomcat:tomcat /opt/tomcat/webapps/sample.war
[root@ngnix tmp]# systemctl restart tomcat
[root@ngnix tmp]# ls -l /opt/tomcat/webapps/
total 16
drwxr-x---. 16 tomcat tomcat 4096 Jun  1 15:16 docs
drwxr-x---.  7 tomcat tomcat  99 Jun  1 15:16 examples
drwxr-x---.  6 tomcat tomcat  79 Jun  1 15:16 host-manager
drwxr-x---.  6 tomcat tomcat 114 Jun  1 15:16 manager
drwxr-x---.  3 tomcat tomcat 4096 Jun  1 15:16 ROOT
drwxr-x---.  5 tomcat tomcat  86 Jun  1 16:03 sample
-rw-r-----. 1 tomcat tomcat 4704 Jun  1 16:03 sample.war
```

Snipping To
Screenshot co

14. Test Tomcat:-

#curl <http://127.0.0.1:8080/sample/>

```
[root@ngnix tmp]# curl http://127.0.0.1:8080/sample/
<html>
<head>
<title>Sample "Hello, World" Application</title>
</head>
<body bgcolor=white>

<table border="0">
<tr>
<td>

</td>
<td>
<h1>Sample "Hello, World" Application</h1>
<p>This is the home page for a sample application used to illustrate the
source directory organization of a web application utilizing the principles
outlined in the Application Developer's Guide.
</td>
</tr>
</table>

<p>To prove that they work, you can execute either of the following links:
<ul>
<li>To a <a href="hello.jsp">JSP page</a>.
<li>To a <a href="hello">servlet</a>.
</ul>

</body>
</html>
[root@ngnix tmp]#
```

15.NGINX Installation:-

```
#dnf install nginx -y
```

```
#systemctl enable --now nginx
```

```
x6 [root@ngnix tmp]# dnf install nginx -y
Updating Subscription Management repositories.
x8 Last metadata expiration check: 1:38:11 ago on Mon 01 Jun 2026 02:32:36 PM IST.
Dependencies resolved.
=====
=====
Package                                Architecture      Version
      Size
=====
=====
Installing:
  nginx                                x86_64            2:1.20.1-28.el9_8.2
      44 k
Installing dependencies:
  nginx-core                          x86_64            2:1.20.1-28.el9_8.2
      579 k
  nginxfilesystem                     noarch            2:1.20.1-28.el9_8.2
      17 k
  redhat-logos-httpd                  noarch            90.6-1.el9
      15 k

Transaction Summary
=====
=====
```

16.Configure NGINX Virtual Hosts

#vi /etc/nginx/conf.d/virtualhost.conf

```
root@nginx:/tmp
[linux 6] [samyak@nginx ~]$ cat /etc/nginx/conf.d/virtualhost.conf
[linux 8] server {
    listen 80;
    server_name www.radical.com;

    location / {
        proxy_pass http://127.0.0.1:8080/sample/;

        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
}

server {
    listen 80;
    server_name www.samyak.com;

    location / {
        proxy_pass http://127.0.0.1:8080/sample/;

        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
}
[samyak@nginx ~]$
```

17.SELinux:-

#getenforce (checks current SELinux mode)

#setsebool -P httpd_can_network_connect 1 (Allow NGINX/Apache (httpd) to connect to network services)

18.Check Nginx status:

#nginx -t

#systemctl restart nginx

19. Firewall Configuration :-

```
#firewall-cmd --permanent --add-service=http
```

```
#firewall-cmd --permanent --add-port=8080/tcp
```

```
#firewall-cmd --reload
```

20. Final Testing On Tomcat Server:-

```
#curl http://127.0.0.1:8080/sample/
```

```
[root@ngnix tmp]# curl http://127.0.0.1:8080/sample/

<html>
<head>
<title>Sample "Hello, World" Application</title>
</head>
<body bgcolor=white>

<table border="0">
<tr>
<td>

</td>
<td>
<h1>Sample "Hello, World" Application</h1>
<p>This is the home page for a sample application used to illustrate the
source directory organization of a web application utilizing the principles
outlined in the Application Developer's Guide.
</td>
</tr>
</table>

<p>To prove that they work, you can execute either of the following links:
<ul>
<li>To a <a href="hello.jsp">JSP page</a>.
<li>To a <a href="hello">servlet</a>.
</ul>

</body>
</html>
[root@ngnix tmp]#
```

21. Final Testing On NGINX Server:-

#curl -H "Host: www.radical.com" <http://127.0.0.1>

```
[root@nginx tmp]# curl -H "Host: www.radical.com" http://127.0.0.1
<html>
<head>
<title>Sample "Hello, World" Application</title>
</head>
<body bgcolor=white>

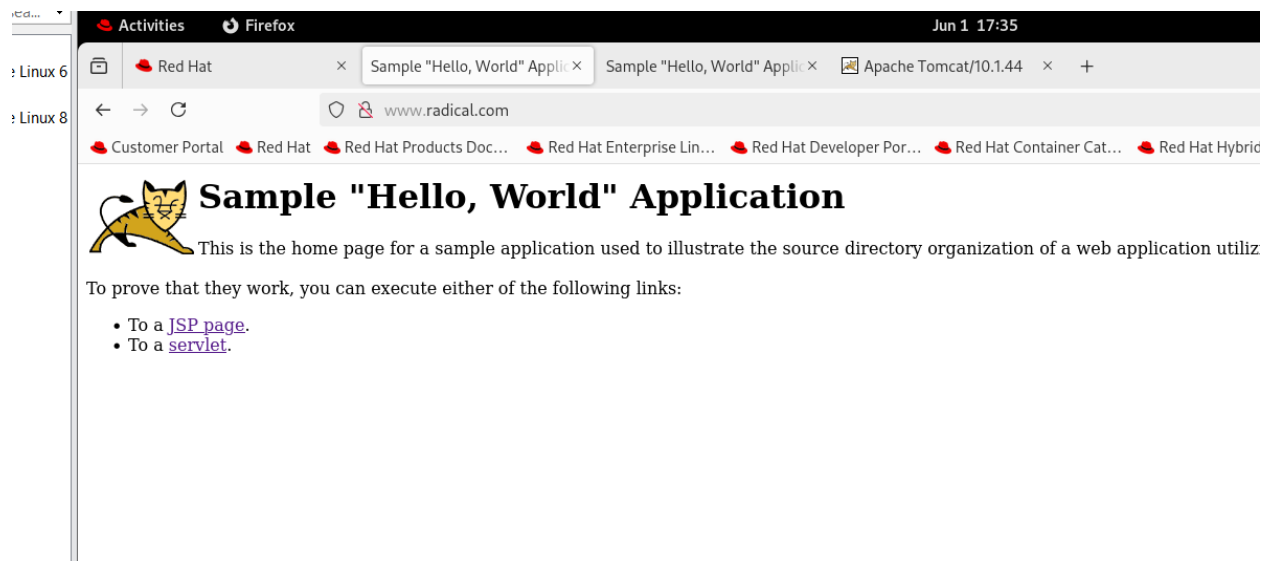
<table border="0">
<tr>
<td>

</td>
<td>
<h1>Sample "Hello, World" Application</h1>
<p>This is the home page for a sample application used to illustrate the
source directory organization of a web application utilizing the principles
outlined in the Application Developer's Guide.
</td>
</tr>
</table>

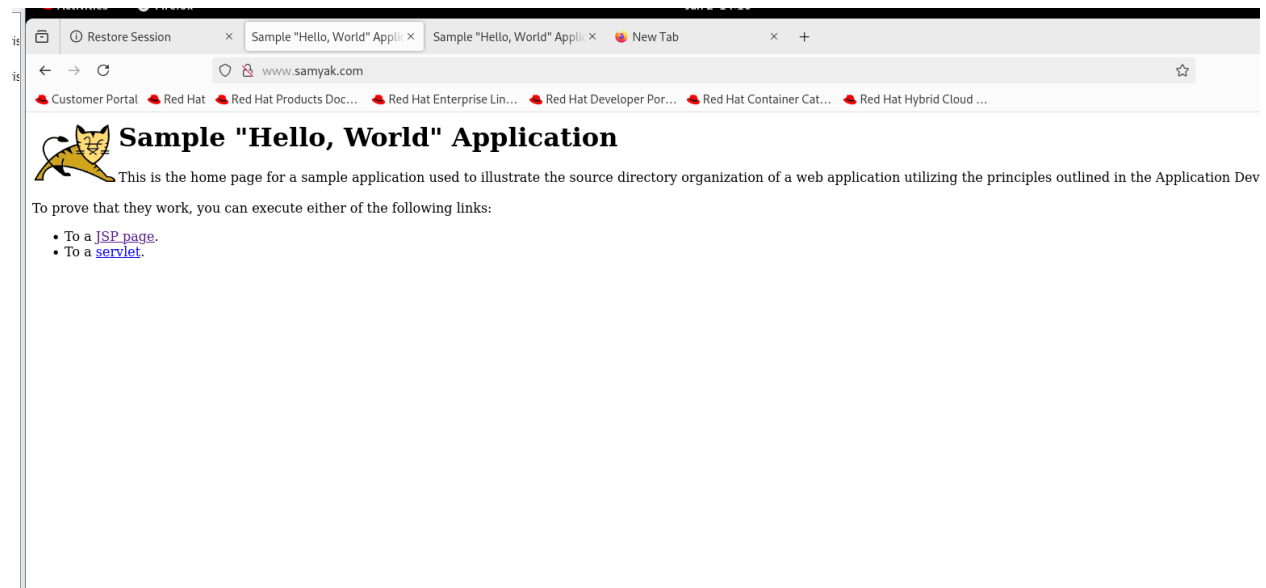
<p>To prove that they work, you can execute either of the following links:
```

22. Open browser:-

<http://www.radical.com>

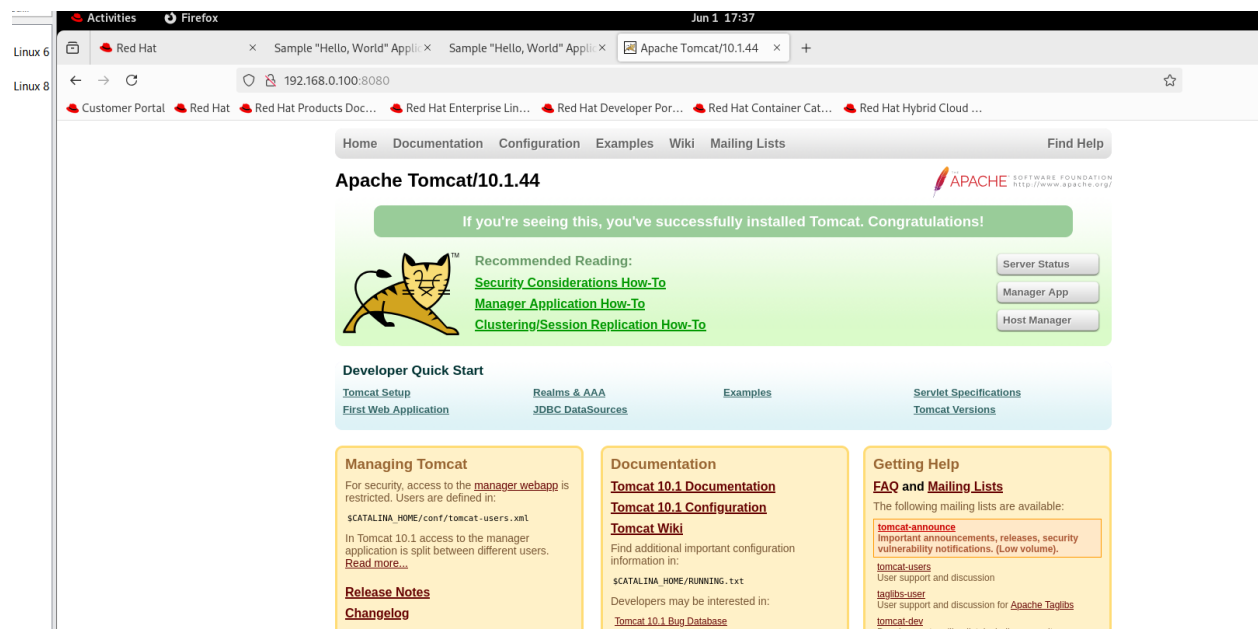


<http://www.samyak.com>



23. Also if you want to check tomcat:-

<http://192.168.0.100:8080>



24.Conclusion

NGINX was successfully installed and configured on RHEL 9 to host multiple websites using Virtual Hosts. DNS records were configured to resolve domain names to the server IP, and firewall settings were updated to allow web traffic.

After testing with tools like curl, ping, and a web browser, all websites were accessible successfully